



Mark Scheme (Results)

November 2025

Pearson Edexcel International GCSE in Chemistry (Modular)
4WCH1/1C

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General Marking Guidance

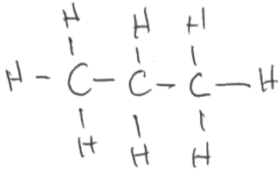
- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Question number	Answer	Notes	Marks
1 (a)	M1 X melting M2 Y boiling M3 Z sublimation	ALLOW evaporation ALLOW vaporisation	3
(b)	$\text{H}_2\text{O (g)} \rightarrow \text{H}_2\text{O (l)}$ M1 both formulae correct M2 both state symbols correct	IGNORE 'water' instead of formulae IGNORE letter case M2 dep on M1	2
(c)	M1 solid (particles) have a regular arrangement OR gas particles are randomly arranged M2 solid (particles) are closely/tightly packed OR gas particles widely spaced	IGNORE statements related to liquids IGNORE diagrams IGNORE statements related to movement including 'random movement'	2
			Total 7

Question number	Answer	Notes	Marks
2 (a)	2 (dyes)	ALLOW two (dyes)	1
(b)	M1 $0.9 \div 6.9$ M2 0.13	ALLOW tolerance so distance to spot in range of 0.8 to 1.0 and distance to solvent front is 6.9 to 7.1 ALLOW ECF from 2 nd dot	2
(c)	M1 the dye is insoluble in water M2 a solvent has to be chosen in which the dye is soluble / solubility in another solvent is more than in water	IGNORE references to speed ALLOW example of a suitable named solvent / to compare the solubility in different solutions	2
			Total 5

Question number			Answer	Notes	Marks
3	(a)	(i)	B (X) <i>A is incorrect as W is a neutron</i> <i>C is incorrect as Y is an electron</i> <i>D is incorrect as Z is the nucleus</i>		1
		(ii)	A (W) <i>B is X incorrect as X is a proton which has a +1 charge</i> <i>C is incorrect as Y is an electron which has a -1 charge</i> <i>D is incorrect as Z is the nucleus which contains protons so has an overall positive charge</i>		1
		(iii)	Nucleus		1
		(iv)	B (3) <i>A is incorrect as it is not 2</i> <i>C is incorrect as it is not 5</i> <i>D is incorrect as it is not 11</i>		1
		(v)	A (2) <i>B is incorrect as it is not 3</i> <i>C is incorrect as it is not 5</i> <i>D is incorrect as it is not 11</i>		1
		(vi)	B		1
	(b)	(positive) ion/cation	ALLOW ‘an ion’ REJECT ‘anion’ REJECT negative	1	
	(c)	(i)	scale on y-axis starting at 6.0 and going up in twos for each large square		1
		(ii)	all points plotted to the nearest grid line	ECF for correct points on an appropriate alternative axis	2
		(iii)	there is a positive correlation (between the relative atomic mass and the atomic number) / as the atomic number increases so does the relative atomic mass	REJECT directly proportional REJECT proportional REJECT linear	1
					Total 11

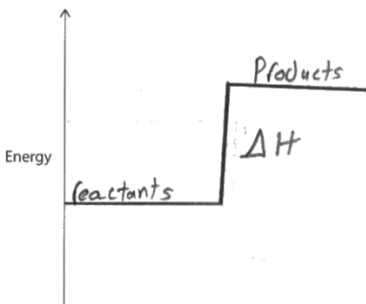
Question number	Answer	Notes	Marks
4	(a)	potassium aluminium iron copper	one mark if both potassium and copper are in the correct order one mark if aluminium and iron are in the correct order
	(b)	M1 the iron is coated with a material M2 which prevents oxygen/water vapour reaching / reacting with the iron	ALLOW examples eg paint, oil, plastic
	(c)	M1 zinc is oxidised, (iron is prevented from being oxidised) M2 zinc loses electrons, (which prevents iron from losing electrons)	REJECT zinc displaces iron IGNORE iron gains electrons so is reduced If no other mark scored, allow one mark for definition of a redox reaction in terms of electrons
	(d)	$\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$ M1 LHS is correct M2 RHS is correct	IGNORE state symbols
			Total 8

Question number	Answer	Notes	Marks
5 (a)	Any five from: M1 use a (fractionating) column to separate the fractions M2 fractions can be separated due to different boiling points M3 crude oil is a mixture of hydrocarbons M4 the mixture is heated M5 molecules with lower boiling points / shorter chains rise to the top of the fractionating column ORA M6 then fractions / molecules condense (and are separated) M7 the fractionating column is cooler at the top / ORA	ALLOW boiled REJECT elements REJECT elements	5
(b)			1
(c)	B (C_nH_{2n+2}) <i>A is incorrect as it is the wrong formula</i> <i>C is incorrect as it has too many hydrogen atoms</i> <i>D is incorrect as it is an alkene</i>		1
(d)	Any two from: M1 carbon M2 carbon monoxide M3 water (vapour)	ACCEPT correct formulae ACCEPT soot ACCEPT steam	2
			Total 9

Question number	Answer	Notes	Marks
6 (a)	A description that refers to six of the following points: M1 hydrochloric acid is transferred into a beaker M2 hydrochloric acid is (gently) warmed M3 zinc oxide is (slowly) added to the acid M4 (zinc oxide is added) until in excess M5 (excess zinc oxide) is filtered M6 solution is gently heated to crystallisation point M7 remaining water is left to evaporate / solution is left to cool M8 crystals patted dry / oven dried	ALLOW suitable alternative wording ALLOW zinc oxide is stirred in the acid	6
(b)	M1 $(3.2 \div 5.5) \times 100$ M2 58(%)	ALLOW answers to any significant figures except 1 sf ALLOW 0.58 for one mark	2
			Total 8

Question number	Answer	Notes	Marks
7 (a) (i)	any value between 21.4 and 21.8 inclusive		1
(ii)	any value between 16.1 and 16.3 inclusive		1
(b) (i)	M1 should not be nitric acid added/sodium hydroxide should be added (not nitric acid) M2 the point where the lines cross shows where the acid was neutralised/the acid was neutralised before 40 cm ³ / the acid was neutralised at 21.6 cm ³	ALLOW any value between 21.4-21.8 cm ³ for 21.6 cm ³	2
(ii)	Any one statement that is: A suitable description e.g. The temperature decreases after the neutralisation point/after 21.6cm ³ (or any value between 21.4-21.8) OR A suitable explanation e.g. The second line shows that there is no further reaction and the cool sodium hydroxide solution decreases the temperature OWTTE		1
(c) (i)	M1 mass of solution = 25 + 25 OR 50 (g) M2 $Q = mc\Delta T$ OR $3780 = 50 \times 4.2 \times \Delta T$ M3 $\Delta T = \frac{3780}{50 \times 4.2}$ OR 18 (°C) M4 (maximum temperature = 19 + 18 =) 37 (°C)	ALLOW ecf for M2 onwards answer of 36 scores 2 answer of 55 scores 3 correct answer without working scores 4	4
(ii)	M1 $\frac{3780}{0.079}$ OR 47 848 (J/mol) M2 47.8 (kJ/mol) M3 – 47.8 (kJ/mol) OR M1 $\frac{3780}{1000}$ OR 3.78 (kJ) M2 $\frac{3.78}{0.079}$ OR 47.8 (kJ/mol) M3 – 47.8 (kJ/mol)	correct answer without working scores 3 or 2 without or incorrect sign Incorrect rounding loses M2	3
			Total 12

Question number	Answer	Notes	Marks
8 (a)	C (magnesium hydroxide) <i>A is incorrect as common ammonium compounds are soluble</i> <i>B is incorrect as lithium chloride is highly soluble</i> <i>D is incorrect as all nitrates are soluble</i>		1
(b)	calcium carbonate / calcium sulfate	IGNORE formulae	1
(c) (i)	M1 Ar values for Mg, O and H identified M2 $24 + (16 \times 2) + (1 \times 2)$ OR $24 + 2(16+1)$ M3 58	ALLOW all 3 marks for answer of 58 ALLOW OH has M_r of 17 ALLOW 2 marks for answer of 41	3
(ii)	OH / hydroxide (ion)	ALLOW OH ⁻	1
(iii)	14	ALLOW any number in the range of 11 to 14	1
(iv)	Mg(OH) ₂ + 2HCl → MgCl ₂ + 2H ₂ O M1 All formulae correct M2 equation is balanced	ALLOW multiples and fractions	2
			Total 9

Question number	Answer	Notes	Marks
9 (a)	B Silver <i>A is incorrect as calcium is too reactive</i> <i>C is incorrect as sodium is too reactive</i> <i>D is incorrect as zinc is too reactive</i>		1
(b)	M1 aluminium is extracted by electrolysis M2 iron is extracted by heating / reduction / displacement with carbon M3 aluminium is more reactive than carbon and iron is less reactive than carbon	ALLOW use of blast furnace ALLOW aluminium is more reactive than iron/ORA	3
(c)	M1 steel is harder / ORA M2 atoms/ions in steel have different sizes / ORA M3 the layers in steel are disrupted OR layers of atoms/ions can't slide over each other / ORA	ALLOW comparatives of stronger / more malleable. ALLOW particles throughout, but REJECT molecules once	3
(d)	 M1 products are higher than reactants M2 reactants and products labelled	ALLOW names or formulae	2
(e)	An explanation that links together the following two points: M1 more energy is taken in / required to break bonds M2 than energy released to make bonds	If state/imply the energy required to make bonds OR if state/imply energy released when bonds are broken scores no mark	2
			Total 11

Question number	Answer	Notes	Marks
10 (a)	burette		1
(b)	Any 5 from: M1 rinse/fill a burette with sulfuric acid M2 use a pipette to add sodium hydroxide solution (to the conical flask) M3 place the conical flask on a white tile M4 add a few drops of indicator (to the conical flask) M5 (add acid and) swirl the contents of the conical flask M6 add (sulfuric) acid dropwise towards the end point (until the indicator changes colour) M7 record initial and final reading on burette	ALLOW any named indicator except universal indicator ALLOW shake IGNORE any colour changes	5
(c)	M1 amount NaOH = $\frac{0.125 \times 25}{1000}$ = 0.003125 (mol) M2 (amount) H₂SO₄ = 2:1 ratio OR 0.0015625 (mol) M3 concentration H₂SO₄ = $\frac{1000 \times 0.0015625}{22.50}$ M4 concentration H₂SO₄ = 0.0694 mol/dm³	Correct answer scores all 4 marks ALLOW ecf from M2 ALLOW any number of significant figures	4
			Total 10